

B.Tech. Degree IV Semester Supplementary Examination April 2025

CE/CS/EC/EE/IT/ME/SE

19-200-0401 COMPLEX VARIABLES AND PARTIAL DIFFERENTIAL EQUATIONS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

CO1: Transform a region to another region using conformal mapping.

CO2: Evaluate real integrals using residue theorem.

CO3: Form and solve partial differential equation.

CO4: Determine solution of partial differential equation for vibrating string and heat conduction.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A

(Answer ALL questions)

(8 × 3 = 24)

	Marks	BL	CO	PO
I. (a) Show that the function $f(z)$ defined by	3	L2	1	3
$f(z) = \frac{x^3(1+i) - y^3(1-i)}{x^2 + y^2} : z \neq 0$				
$= 0 : z = 0$				
satisfies CR equations. But not analytic at the origin.				
(b) Show that the real and imaginary parts of an analytic function are harmonic.	3	L1	1	4
(c) Evaluate $\int_0^{1+i} (x^2 - iy) dz$ along the parabola $y = x^2$.	3	L2	2	4
(d) Using Cauchy's integral formula, evaluate	3	L3	2	3
$\int_c \frac{e^{-2z}}{(z-1)^3} dz$, where c is $ z = 2$.				
(e) Form the partial differential equation by eliminating arbitrary constants from $(x-a)^2 + (y-b)^2 + z^2 = 1$.	3	L2	3	3
(f) Solve $p^2 + q^2 = 4pq$.	3	L5	3	4
(g) By the method of separation of variables, solve $U_{xx} - U_y = 0$.	3	L3	4	4
(h) Find the solution of one-dimensional wave equation	3	L1	4	3

PART B

(4 × 12 = 48)

II. (a) Find the analytic function $f(z)$ whose real part is $e^{2x}(x \cos 2y - y \sin 2y)$.	6	L1	1	2
(b) An electrostatic field in the xy-plane is given by the potential function $\phi = 3x^2y - y^3$. Find the stream function.	6	L2	1	4
OR				
III. (a) Find the Bilinear transformation which maps the points $z = 1, i, -1$ on the points $w = i, 0, -i$.	6	L2	1	4
(b) Prove that the transformation $w = \frac{1}{z}$ maps a circle onto a circle or a straight line if the former passes through origin.	6	L5	1	3

(P.T.O.)

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		Marks	BL	CO	PO
IV.	(a) Verify Cauchy's theorem $f(z) = z + 1$ and c is the boundary of a square whose vertices are at the points $z = 0, z = 1, z = 1 + i, z = i$.	6	L3	2	4
	(b) Expand $\frac{1}{z^2 - 3z + 2}$ in the region $1 < z < 2$.	6	L5	2	3
	OR				
V.	(a) Using Residue Theorem evaluate $\int_c \frac{2z-1}{z(z-1)(z-3)} dz$ where c is $ z = 2$.	6	L3	2	4
	(b) Evaluate $\int_0^{2\pi} \frac{d\theta}{a + b \cos \theta}$ where $a > b $.	6	L5	2	4
VI.	(a) Solve $x(y-z)p + y(z-x)q = z(x-y)$.	6	L4	3	4
	(b) Solve $p^2 + q^2 = x^2 + y^2$.	6	L4	3	3
	OR				
VII.	(a) Solve $z^2 = 1 + p^2 + q^2$.	6	L4	3	4
	(b) Solve $(D^3 - 7DD'^2 - 6D'^3)z = \sin(x + 2y)$.	6	L4	3	4
VIII.	(a) Derive the wave equation.	6	L1	4	4
	(b) A string is stretched and fastened to two points l apart. Motion is started by displacing in the form $y = a \sin \frac{\pi x}{l}$ from which it is released at time $t = 0$. Show that the displacement of any point x from one end at time t is given by $y(x, t) = a \sin \frac{\pi x}{l} \cos \frac{\pi ct}{l}$.	6	L4	4	4
	OR				
IX.	(a) Derive one dimensional heat equation.	6	L1	4	4
	(b) Solve Laplace's equation by the method of separation of variables.	6	L4	4	4

Blooms's Taxonomy Level's

L1 - 21%, L2 - 21%, L3 - 17%, L4 - 25%, L5 - 16%.
